Fiscal Year:	FY 2021	**Task Last Updated:**	FY 10/16/2021
PI Name:	Azzam, Edouard Ph.D.		
Project Title:	Oxidative Stress and the Cancer Risk of Space Radiation		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Key Personnel Changes/Previous PI:	NOTE: Principal Investigator Edouard Azzam, Ph.D., retired in early 2021 and CoI Roger Howell, Ph.D., was named PI for university purposes only. Dr. Azzam remains as principal researcher.		
COI Name (Institution):	de Toledo, Sonia Ph.D. (Rutgers University, New Jersey Medical School) Howell, Roger Ph.D. (Rutgers University, New Jersey Medical School) Pain, Debkumar Ph.D. (Rutgers University, New Jersey Medical School)		
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Task Description:	The objective of this project is to investigate transient and persistent oxidative stress, and its association with cancer induction, after exposure of mice to low doses/fluences of different types of space radiation. The proposal is based on the hypothesis that space radiations with different biophysical properties induce distinct redox-modulated biochemical changes. Such changes may differentially perturb physiological functions and may induce DNA damage to different extents. If they persist, some of these changes may lead to cancer. This is an immediate concern to NASA, particularly in the context of long-duration exploratory space missions. This proposal will use middle-aged mice to determine the effects of space radiation on critical redox-modulated cellular processes. Experiments will include exposures to low doses of different high energy particles (oxygen, calcium, and silicon), delivered at low dose-rate. The results will be compared with those obtained in mice exposed in parallel to cesium-137 gamma rays. We will examine acute and chronic oxidative changes in DNA, and in lipids and proteins involved in critical signaling pathways that mediate the cellular responses to stress. We will measure these changes in radiation sensitive and resistant organs following whole or partial body irradiation of mice strains that vary in their susceptibility to cancer. We will also investigate stressful effects in irradiated organs/tissues and their propagation to non-irradiated organs/tissues. We will explore the possibility that prior exposure to high energy protons induces mechanisms that protect tissues from the targeted and non-targeted stresses due to a subsequent exposure to low fluences of highly damaging energetic particles. The goal is to generate data related to Specific Gaps in knowledge listed in Cancer 1-Cancer 5 and in Cancer-7, which may help reduce the uncertainty in estimating cancer risk to astronauts.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	There is overwhelming evidence to support that oxidative stress contributes to elevated levels of DNA damage, abnormal growth control, and altered metabolic pathways, which can lead to cancer. However, the effects of space ionizing radiation (IR) on these processes in vivo and the underlying signaling events have not been identified, particularly in the context of chronic exposure to low fluences of energetic high atomic number and high energy (HZE) particles that vary in their linear energy transfer (LET). The issue is further complicated by the fact that astronauts are exposed to mixed types of IR. An exposure to a low dose of low-LET IR prior to a dose from high-LET IR may induce protective processes that attenuate the damaging effects of the latter. This is important because the low flux of the high-LET HZE radiations in space relative to the higher flux of low-LET protons makes it highly probable that for any given cell in the body, proton events will precede any HZE event. Assessing these targeted and non- targeted responses will synergize with other NASA supported studies and will contribute crucial and novel mechanistic information to ongoing efforts in developing biophysical models for predicting health risks to astronauts. By achieving an integrated understanding of the endpoints investigated in this proposal, a rational path towards preventing the occurrence or delaying the onset of cancer (and other adverse health effects) during or after space missions may be developed. Further, as particle therapy is being increasingly used to treat cancer, the proposed studies may lead to the development of treatment protocols that enhance the efficacy of anti-tumor treatments and attenuate post therapeutic out-of-field normal tissue toxicity.
	This project tested the central hypothesis that exposures to and high atomic number, high energy (HZE) particles induce distinct redox-modulated biochemical changes that perturb essential signaling networks leading to persistent oxidative stress and progression towards malignancy in both irradiated and non-irradiated tissues. Further, these effects depend on the linear energy transfer (LET) of the radiation and are mitigated by prior exposure to protons. The experimental design relied on exposing groups of middle-aged CBA/CaJ male mice (9–10-month-old, to simulate age of active astronauts) to low/moderate mean absorbed doses of isovelocity 1 GeV/u protons, oxygen, silicon, or calcium ions, with respective average LET values of ~0.24, 14, 44, and 88 keV/µm. The radiation was delivered to the whole body or head only. A different set of mice were exposed to 137Cs gamma rays as reference radiation. Control mice were sham treated in parallel. Various redox modulated changes, inflammatory responses and histological structural changes were examined as a function of time after irradiation (2 weeks and up to 18 months). Cancer incidence was also evaluated in targeted and non-targeted organs. The groups of mice were as follows: 1: Control; 2: gamma rays: 1.5 Gy (acute, single fraction, whole body); 3: gamma rays: 3 Gy (acute, single fraction, whole body); 4: 1 GeV protons: 0.2 Gy (0.0035 Gy/min, whole body); 5: 1 GeV/u Ca: 0.2 Gy (in 3 fractions; 1 acute fraction/day; whole body); 6: 1 GeV/u Ca: 0.3 Gy (in 3 fractions; 1 acute fraction/day; whole body); 7: 1 GeV/u Ca: 0.4 Gy (in 3 fractions; 1 acute fraction/day; whole body); 8: 1 GeV/u Ca: 0.4 Gy (in 1 acute fraction; whole body); 9: 1 GeV/u Ca: 0.4 Gy (in 1 acute fraction; head only); 10: 1 GeV/u Si: 0.4 Gy (in 3 fractions; 1 acute fraction/day; whole body); 11: 1 GeV/u Si: 0.4 Gy (in 1 acute single fraction; whole body); 12: 1 GeV/u O: 0.4 Gy (in 3 fractions; 1 acute fraction/day; whole body); 13: 1 GeV/u O: 0.4 Gy (in 1 acute single fraction; whole body); 14: 1 GeV protons followed by 1 GeV/u Ca: whole body exposure to 0.2 Gy of protons delivered 24 h prior to 0.4 Gy of 1 GeV/u Ca ions delivered to the whole body; 15: 1 GeV protons followed by 1 GeV/u Ca: whole body exposure to 0.2 Gy of protons delivered 24 h prior to 0.4 Gy of 1 GeV/u Ca ions targeted to the head only. Major highlights: - Effects on modulation of the abundance of circulating hematopoietic cells and bone marrow stem cells. Up to 10-fold increases in circulating neutrophils were detected at two weeks in mice exposed to 20, 30, or 40 cGy of either of the heavy ions delivered in a fractionated manner ($p<0.001$). The groups also showed an increase in circulating monocytes ($p<0.01$). The mice exposed to a single bolus of 40 cGy of Ca ions did not show significant increases at 2 weeks; however, by 3 months, increases in neutrophils were detected ($p<0.001$). These increases in neutrophils and monocytes in circulating blood were associated with decreases in these cell subsets in bone marrow ($p<0.05$), suggesting mobilization out of this compartment. Common myeloid, as well as granulocyte / macrophage and megakaryocyte-erythroid progenitors were decreased ($p<0.1$) in bone marrow. Notably, decreases ($p<0.01$) in short-term hematopoietic stem cells were detected. The alterations observed at 2 weeks were associated with changes in the levels of circulating inflammatory cytokines. Furthermore, histological analyses revealed prominent interstitial lung disease in mice exposed to a single bolus of 40 cGy of heavy ions, characterized with thickened alveolar septa, pulmonary congestion, and endothelial hyperplasia. The mice exposed to the fractionated regimens presented mild lung injury. The early response of neutrophils and monocytes in mice exposed to the energetic heavy ions returned to a normal range at 6 months after irradiation and remained in this normal range at 13 months. However, at the latter time point, the proportion (%) of circulating plasma cells, but not B cells, were increased ($p<0.001$) in mice exposed to 30 or 40 cGy of Ca ions delivered in a fractionated manner. The effect seems to depend on the radiation dose and delivery manner. Similar finding occurred in mice exposed to 40 cGy of silicon or oxygen ions, but not in mice exposed to gamma rays or

Task Progress:	protons. This indicates the phenotypes may be specific to high LET radiations and is suggestive of the development of a plasma cell dyscresia. At 15-18 months after exposure to HZE particles, an impaired cytotoxic function of natural killer (NK) cells was also observed, which suggests long-term overall suppression and decreased function of these cells. There was no change in the number of red blood cells or the concentration of hemoglobin across all irradiated groups; however, there was an increase in the hematocrit percentages for animals irradiated with 40 cGy of 1 GeV/u Ca ions delivered as a single bolus to the whole body ($p = 0.03$). The blood of the irradiated mice was also analyzed for changes in triglycerides, cholesterol content, as well as concentration of several biomarkers. - Long-term effects on morphological changes and radio-densitometric characteristics of bone. Consistent with an effect on bone marrow stem cells, computed tomography (micro-CT) revealed morphological change in trabecular bone at distal femur metaphysis at two weeks after exposure to the HZE ions. Trabecular bone in these mice demonstrated more rod shaped, instead of mostly plate-shaped in the control mice. In addition, CT and micro-CT performed at 15-18 months after irradiation revealed persistent effects on bone density. Intra-group co-plots showed that the normalized bone density distributions for the mice in the 0 cGy (control) group exhibit a high degree of overlap, displaying limited variation throughout the distributions. In contrast, intra-group co-plots of the data for the heavy ion and gamma-irradiated mice revealed marked distribution heterogeneity within these test groups. Micro-CT examinations were consistent with aberrant mineralization in HZE-irradiated mice. - Ultrasound echocardiography (effect on heart physiology). Relative to gamma-irradiated mice, there were significant changes ($p<0.05$) in cardiac output of mice irradiated with the higher doses of HZE particles (30, 40 cGy). A change in the left ventricular mass index was also observed ($p<0.02$). - Oxidative stress and inflammation in various organs. Oxidative stress in various tissues was detected by oxyblot assay. Analyses of fibrosis, at 15-18 months after exposure to HZE particles, in lung, liver, kidney, and heart, revealed a significant persistence of inflammation ($p<0.01$ to $p<0.001$). The fibrosis was associated with increases in the levels of proteins harboring 3-nitrotyrosine and lipid peroxidation adducts (4-hydroxynonenal). Analyses of mitochondrial stress in heart tissue was also performed. The results showed significant stress at early times after irradiation. The induced stress was significantly attenuated by 15-18 months after exposure. At 15 months, antioxidant enzyme analyses in kidney (superoxide dismutases, catalase, glutathione-S-transferase) revealed a trend of increased activity, being significant for catalase ($p<0.05$). The increases, however, were not sufficient to eliminate the induced oxidative stress. H&E (Hematoxylin and Eosin) staining showed structural changes in kidney, liver, and lung. Analyses of structural changes in brain are ongoing. Also, analyses have been performed to investigate microgliosis and astrogliosis in all the treated groups. These analyses are near completion. Analyses of signaling pathways associated with DNA damage, oxidative stress, and inflammation were pursued in various organs. - Pathological changes in reproductive tissues. At 15-18 months after irradiation, notable increases ($p<0.0001$) were observed in the size of the seminal vesicle during gross examination of mice exposed to 40 cGy of 1 GeV/u of Ca ions. In contrast, the size of the seminal vesicles in mice exposed to 150 cGy of gamma rays were similar to those observed in control mice. - Global changes in protein expression level and S-nitrosylation. To understand the effects of whole or partial body exposure to space radiation on signal transduction mediated by S-nitrosylation, we performed mass spectrometry analyses on the cerebellum of sham-irradiated mice or cerebellum or mice that received whole body or head only exposures to 40 cGy from 1 GeV/u of calcium ions delivered acutely as a single bolus. The mice used in experiments were sacrificed 14 days after irradiation. Promptly after harvesting the cerebellums, the proteins were extracted and processed for analyses. The resulting expression patterns of proteins (general proteome) and S-nitrosylated protein (S-nitroso proteome) were analyzed by bioinformatics classification data mining tools and pathway analysis tools. The results from the protein expression studies indicate that there was no change in protein expression level of neuronal Nitric Oxide Synthase (nNOS) before and after irradiation with 40 cGy of calcium ions targeted either to the whole body or head only. The experiment did not detect the expression of the other NOSs (i.e., inducible and endothelial Nitric Oxide Synthases). Thus, the changes observed in nitrosylated protein cannot be attributed to the change in the level of NOSs. This result does not exclude a possibility where the irradiation may have affected the activity level of NOSs enzymes. However, the changes in nitrosylation observed may be due to alternative mechanisms such as transnitrosylation. - Histological determinations of cancer incidence. Analyses of cancer incidence is in progress with the assistance of a biostatistician. Histological examinations have so far shown that relative to control, cancer incidence in middle-aged mice (9-10 months) is significantly lower than reported by others in mice exposed to HZE particles at younger age.
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